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METHODS OF EXAMINING SOUNDNESS

Presented by

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❑ The following notes serve as an introduction to the examination of animals for soundness.

A- EXAMINATION OF HORSES

a. In the stable:

1- The horse should be observed after having been rested for an hour or more in the stall or loose box to determine evidence of such **stable vices** as **wind- sucking, crib-biting, weaving, rubbing the tail against the walls, etc.....**

2- The **preliminary rest** is important, since many forms of **lameness**, especially those due to **lesions** involving **joints, ligaments, and tendons**, become **intensified after a rest**.

3- In certain instances, it may be desirable to **exercise** a horse or make it work for an hour or two, **then examine for lameness**.

4- It is desirable to **offer a small feed of corn**, firstly to see whether the horse **eats properly** and secondly to ascertain whether it lets the **food fall from the sides of the mouth**.

5- **Urination** and **defecation** should **pass normally**.

6- The **amount** and **type** of **saliva** must be observed; excess saliva, **tears**, or **any abnormal discharge** from the openings of the body indicates unsoundness.

7- The animal should **show interest in the things around him**, such as **moving the ears** forward and backward or to the sides when hearing loud voices.

8- After training, the **sweat** must be **thin, watery**, and **little in quantity**. It will first appear between the ears and on the flanks. In a **diseased animal**, the sweat is **profuse** and **soapy**.

b. Standing still:

- 1-** When the horse is taken outdoors, great care should be taken to observe **how it backs and turns.**
 - 2-** The **head, ears, nostrils, shoulders, forelimbs, and hoof** can be noted.
 - 3-** The observer should move around to stand immediately **behind the horse** and should once again compare the **right and left sides** of the hips, buttocks, gaskin, hocks, hind cannon, and hoof, **noting the degree of symmetry** shown.
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c. Walking and Trotting:

From behind: The groom leading the horse on the left side should be instructed to walk the horse straight ahead, and the first steps should be carefully observed.

- 1- whether **each foot comes freely off the ground.**
- 2- whether the **muscles play their part freely,**
- 3- whether **each foot** is lifted and carried forward on **equal distance.**
- 4 whether **the same amount of shoe is shown.**
- 5- whether **the sound made by putting the foot down** is approximately the same with each foot.

From in front: The horse should then be walked straight toward the observer so that the action can be observed from the front as the horse approaches.

The observer should step aside and watch the horse walk by from both sides **to see how much the limb muscles move.**

Subsequently, the horse should be **trotted** in the same direction as for the walk examination.

d. Manual Examinations:

- 1- Commence at the **head**, noting the **color of the mucous membranes of the eyes and mouth**, the **state of the teeth**, and pass the hand over the **poll** under the **jowl**.
- 2- Standing on the **left side**, pass the left hand down the **neck** over the **shoulders, withers**, and then run down at the left **elbow, fore knee, back tendons**, and **feet** for **bony enlargements or thickening of tendons**, and finally pick up the **left fore foot**.
- 4- Facing the **rear**, the hand should be run over the **horse's back** to the **hind limb**, which is **examined in the same way as the fore limb**.

5- The **shoes** should be **removed** for careful examination of the **sole, frog, bars, and lower parts of the wall.**

6-The tail should be lifted and the **anus, vulva, or penis and scrotum** examined.

7- The **coat** must be **shiny, fine, and not hard**, and the **skin** should be **freely movable**, and the skin must be **free from ectoparasites.**

e. Examination of the particular parts:

1- Particular examination must be made of the **eyes**, the **windpipe**, and the **heart**.

2- All the **mucous membranes** must be **moistened** and have a **rosy red color** and be free from any spots or discharges. If they are **congested**, this indicates **fever**; **pale** coloration indicates **anemia**; **yellow** coloration indicates **liver disorders**; **cyanosis** indicates **toxicity**; and **petechial hemorrhages** indicate **blood parasites**.

3- In **stallions** to be used for **breeding**, a further examination of the **penis**, **testes**, **accessory sexual glands**, and the **semen**.

➤ The **testicles** should be approximately **equal size**; they should be **carried at the same level** or one (**right**) may be **carried a little higher** in the **scrotum** than the other.

- When they are palpated, they should show **no signs of pain, hardness, or swelling** either in the **glands themselves** or in the **scrotum**.
- They should be **freely movable inside the scrotum**; any adhesions denote previous inflammation. They should be of **glandular texture**, and the **groove between the testicles and epididymis** should be palpable.
- Collection of samples of **semen** is now necessary to be examined microscopically for **sperm motility, sperm count**, presence of any **abnormalities**, and **viability** of sperm on storage at about 5 °C.
- Stallion must necessarily be **fertile**, possessing **no anatomical, physiological, or histological abnormalities** that will prevent him from holding offspring.
- The stallion must be **free from venereal diseases**.

4- In **mares** to be used for **breeding**:

- examination of the **vagina and cervix** by speculum and the **ovaries and uterus** by rectal palpation or ultrasound examination.
- The mare must necessarily be **fertile**, possessing **no anatomical, physiological, or histological abnormalities** that will prevent her from holding offspring.
- The mare must be free **from venereal diseases**.

f. The age of the animal:

- should be determined by examination of **teeth**.

B- EXAMINATION OF CATTLE

- There is no such corresponding detailed examination of cattle as carried out for horses.
- Bulls & cows should be examined for **general health** and **fitness** before purchase or when exposing them for sale. Animals for export must also be examined and a certificate of health granted before they are accepted for shipment.

a. Examination of bulls must be as follows:

1. The testicles:

- should be approximately **equal size**.
- should be **carried at the same level**, or one (**right**) may be carried a little **higher** in the scrotum than the other.
- When they are **palpated**, they should show **no signs of pain, hardness, or swelling** either in the **glands** themselves or in the **scrotum**.
- They should be **freely movable** inside the scrotum; any adhesions denote previous inflammation.
- They should be of **glandular texture**, and the **groove between the testicles and epididymis** should be palpable.
- In **older bulls**, the **testicles** are generally **carried low**, i.e., will down the line of the belly, but in **cold weather** or during **anger or excitement**, they may be **retracted by the cremaster muscles**.

2. A bull must necessarily be **fertile**, possessing **no anatomical, physiological, or histological abnormalities** that will prevent him from holding offspring.
3. Collection of samples of **semen** is now necessary to be examined macroscopically for **sperm motility, sperm count**, presence of any **abnormalities**, and **viability** of sperm on storage at about 5 °C.
4. The bull must be **free from venereal diseases**.
5. The feet of bull, especially the **hind foot**, are **important**, since **lameness** due to **overgrown or broken horn, foul in the foot**, etc., may **interfere** with the **ability** or willingness to **serve cows successfully**.

b. Examination of cows must be as follows:

1. The **udder** is generally the most important organ to which attention should be paid. It should be **free from hard lumps, firm but not fleshy**. The texture of the udder should be **glandular and uniform, freely pliable after milking**, and quite **painless** when handled.
2. The **teats** must be of **equal size (average length 5-7 cm, average diameter 2.5 - 3.5 cm)**, **neither tall nor short**, and should be **free from nodules** in the inside of the teat canal, which may interfere with the flow of milk; the **external surface** should be **free from warts or abraded areas**.
3. A little milk should be taken from each quarter to ensure that the **teat sphincters** are functioning normally, and each **teat** should **show only one stream of milk**.

4. Any change of **milk** color, odor, quantity, and consistency should be noticed rapidly
5. A **high nervous cow** or one that **kicks** when being **milked** is usually not an advisable purchase for a dairy herd.
6. Each examination will be concerned with ensuring that the **cow is free from diseases**, especially **tuberculosis, John's disease, contagious abortion, and mastitis.**
7. Examine the **reproductive system** of a cow with a speculum, rectal palpation, and ultrasound examination (should be **free from abnormal discharges and inflammation**)

TEAT WARTS



C- EXAMINATION OF SHEEP

1. The sheep of any age should have a clean, **clear eye**, **ears** should be constantly **moved**, and the **head** and **neck** should be held **upright**. At the **walk**, there should be **good coordination** of movement.
2. Other points to look for in healthy sheep are:
 - a. Good **appetite**.
 - b. regular **rumination**.
 - c. **fleece** in the unshorn sheep should be healthy and not easily detached.
 - d. Keeping within the **flock**.
 - e. Sound **feet**.
 - f. Sound unbroken **teeth**.

3. **When a sheep is handled**, it should feel **firm** and **strong**; an animal that feels **limp** is never healthy.

4. In **hot and dry weather**, the **respiration** of **healthy** sheep, even when at **rest**, is much **faster** than normal.

5. The wool of **fine-wool breeds** is **greasy**, but the **coarse-wool breeds** have naturally **dry** and **harsh** wool.

D - EXAMINATION OF GOAT

It takes a great deal of experience and study to be able to tell a good goat from a poor one. To the homesteader, a good goat is simply one that milks well for a reasonably long lactation.

The general appearance of livestock, the way it's put together, is called conformation. These are some of the points of conformation goat breeders look for.

- 1. Head:** should be moderately long with a **convex** or **straight nasal bridge**.
- 2. Ears:** should be **short** and pointed **forward** or **long** and directed **downward**.

3. Muzzle: must be **broad** with **strong jaws**, which indicate **feeding** ability.

4. Nostrils: Must be **large, well-distended** for proper **breathing**.

5. Neck: should be **clean-cut** and **feminine** in the **doe**, **masculine** in the **buck**, with a **length** appropriate to the size of the animal. It must join at the withers with **no "ewe neck"**. The goat needs a **large, well-developed windpipe**.

6. The forelegs: must be set squarely to support the body and **wide apart** to **give room to the chest**.



7. The rib cage and back: the space between ribs should be wide. The **chest** should be broad and deep, indicating a strong respiratory system. The back should be nearly straight with just a slight rise in front of the hip bones.

8. The rump: The slope of the rump should be slight, and the rump should be broad.

9. The barrel: should be large in depth, length, and breadth. A large barrel indicates a large, well-developed rumen, which is necessary for top production.



10. Udder:

- **Abnormalities** such as **extra teats**, **spur teat**, or **teats with double orifices** should be avoided. (In very young kids, **extra teats** can be removed surgically).
- **Very large (Sausage teats)** are undesirable; **very small** ones may be worse as they make **milking difficult**, and take a **longer time**.
- The **well-attached udder**, carried high out of harm's way with **average-sized teats** (average length 2-5 cm, average diameter 1.5 - 2.5 cm) and **free from lumps** and **other deformities**, is the heart of your homestead dairy.

11. Skin: It should be **soft** and **loose** over the barrel. It should be **free from ectoparasites** such as mites and lice.

12. Wattles: Some goats have wattles, which are **small appendages of skin** on the **neck**. Some breeders **remove** from young kids, not only to make the animal look smoother, but also because sometimes another kid will **suck** on them and make them **sore**.

13. Horns: It is common practice to **remove** them at birth because they can cause problems later, as they are **dangerous** to both **other goats** and **their keeper**.

E- EXAMINATION OF CAMEL

1. The type of camel will depend greatly upon the nature of the country in which it will work. Camels bred in **sandy deserts** do not work well in **fields**, and vice versa.

2. Weight carrying capacity and endurance:

- a.** The camel must be capable of carrying a standard load weighing about **250 kg** with force and without laziness or trembling.
- b.** The **more compact** the camel, the **more suitable to carry weights** regardless of its height.
- c.** A **short-legged, short-backed, deep chest** with **well-developed muscles** and **neck** should always be selected in purchasing.
- d.** In **purchasing camels**, **age** must be **at least 5 years**, and the **hump** should be **fully developed** and **plump with fat**.

3. General condition:

- a.** The camel **coat** should be **fine**, **not harsh**, and **not easily detached**, and the **skin** should be **freely movable**.
- b.** The **coat** should **not** show any **bare areas** or **crusts**, which may indicate **skin diseases**, especially **mange**.
- c.** **Healthy** camels should **eat their food** in front of them, but in **ill health**, they do **not care** about their **food**.
- d.** The **feces** of the camel consist of small oval balls, sometimes **bigger than a pigeon's egg**, and **hard** in **consistency**. When the camel is **constipated**, the balls are **harder** and **more elongated** with **twisted ends**.

4. The way the camel sits down:

- a.** The operator must notice the act of sitting down (**kneeling**). If the animal **trembles**, you must **reject** it before purchasing due to its **weakness**.
- b.** If the camel is suffering from **bruised pads** that have been plastered by mud or sand to deceive the purchaser, therefore they must be well **detected** by **observing the way the camel sits down**.
- c.** A **healthy camel sits down straight and evenly**, but when suffering from **bruised pads**, it will not bring its weight down onto the pads, but will **sit down obliquely to avoid the pressure of the painful part**.

5. Free from wounds, malformations, and diseases:

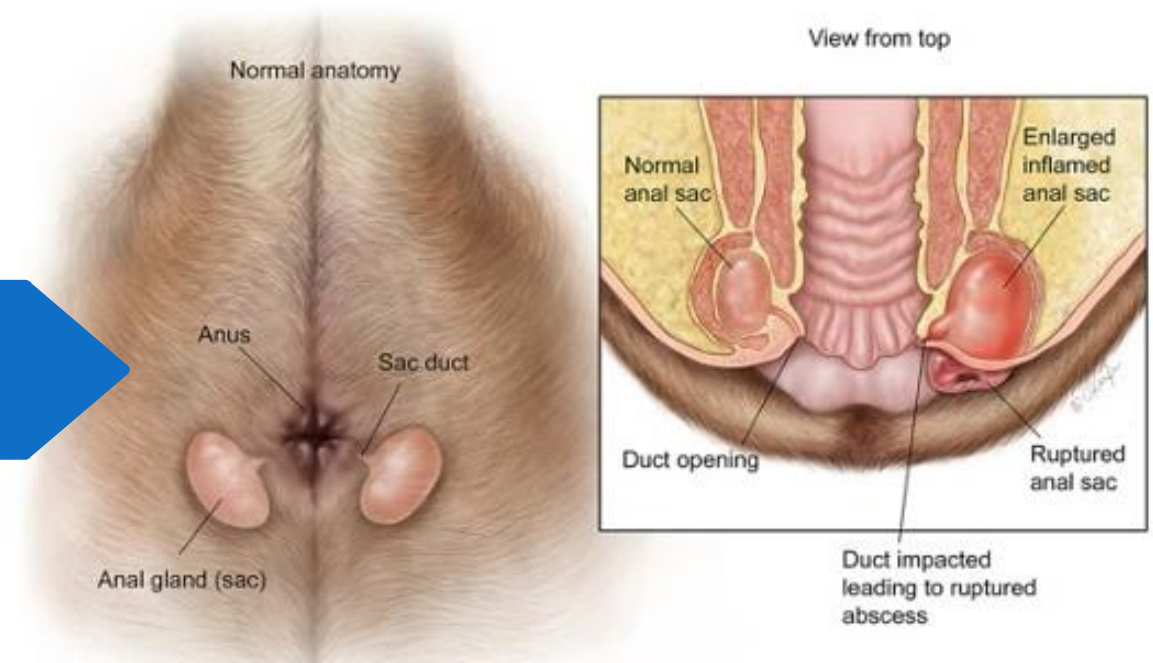
- a. The **most serious wounds** are those of the **foot pads**, **chest pad**, and on that portion of the **back** where the saddle rests.
- b. The **nostrils** should be **free from wounds**.
- c. The **eyes** should be **clear**, bright, and vigorous **without watery discharge** and with **long fringed lashes**.
- d. The animal should be **free from dislocation of the joints**, **lameness**, and **swelling of the lymph nodes**.
- e. Special **microscopical examination** should be applied to the **feces**, **urine**, and **blood**.

F- EXAMINATION OF DOGS

- 1.** All **well-exercised dogs** should be "**muscled up**" and hard, and the **skin** should in all cases be **freely movable**.
- 2.** The **coat** in both **long** and **short** hair **breeds** should be **glossy**, with **no bare patches**.
- 3.** The **breath** should **not** be **offensive**, and the **tongue** should be a healthy **pink**.
- 4.** The **eyelids** should **not** be **thickened**, and the **tips of the ears** should be **free from scars**, which may **not readily heal**.

5. A **moist nose** is usually considered a **sign of good health**.
6. The **anal ring** should be **normal**, i.e. **not thickened**, which may denote **inflamed anal glands**.
7. There should be **no scars** or **thickening between the toes**.
8. A healthy dog will **sleep**, or **rest**, for about **fifteen** of the twenty- four hours.

inflamed anal glands.



9. The **teeth** of a dog should be **strong, sharp, not broken, and white**.

10. **Abnormal discharges** from **genital organs** are signs of **disorder**, though **healthy dogs** often have a **drop of mucous** at the **end** of the **prepuce**.

11. A large percentage of apparently healthy dogs harbor **internal parasites**, so they are a potential **source of ill health**, and it is better to **get rid** of these **parasites** with suitable **anthelmintics**.

G- EXAMINATION OF POULTRY

1. The bird should be **actively vigorous**, scratching and having a great vitality in running from one place to another.
2. The bird should **walk freely** without **dragging** their legs or showing any signs of **lameness**.
3. The healthy bird should **not** be **isolated** alone in any **corner** of the farm but **must** always be **within** the whole **flock**.
4. By palpation of the **joints**, they should **not** be **inflamed** or **swollen**, **freely flexed**, and have **no gouty deposits**.

gouty deposits



5. The **body** should be **free** from **ecto-parasites**, e.g., mites.
6. The bird must be examined by **lifting** it from its **wing** and **comparing** its **size** with its **weight**.
7. The healthy **bird** should be **well covered with muscles**, especially the breast muscles around the keel bone. If **birds have a huge size** and **light weights**, they might denote affections with **spirochaetosis** or **avian tuberculosis**.
8. The bird should have a **bright, prominent eye** without discharges and a **short, stout beak**.

9. The head of the bird should bear a **full red** waxy **comb** and **wattles**. A **pale comb** or **wattles** denotes **anemia**.

10. The **head** must be **normal** in shape, free from **inflammation** or **swelling**, which may denote **avian pox**, **influenza**, and **Newcastle disease**.

11. The **buccal mucous membrane** must be rosy red in color, glistening without **discharges** or **adhesive whitish membrane**, which may denote **diphtheria infections (fowl pox)**.

12. The birds should have a **bloomy**, unruffled, oily **feather**, having **no external parasites** e.g. lice, mites.

Swelling and inflammation of the head of the chicken



**adhesive whitish
mucous membrane,
which may denote
diphtheria infection
(fowl pox).**



13. The **conformation** of a **good layer** is a **broad and long back**, **good depth** of the **body**, and a **long keel bone**.

14. The **body** of a **good laying bird** should **not** be either **too coarse** nor **too fine**, but up to the **standard weight** of the breed.

15. The **laying** bird should have **soft** and **pliable skin**.

THANK
YOU